

Multiple and Nonparametric Regression

2.1 Introduction

In this chapter we discuss some popular linear methods for regression analysis with continuous response variable. We call them linear regression models in general, but our discussion is not limited to the classical multiple linear regression. They are extended to multivariate nonparametric regression via the kernel trick. We first give a brief introduction to multiple linear regression and least-squares, presenting the basic and important ideas such as inferential results, Box-Cox transformation and basis expansion. We then discuss linear methods based on regularized least-squares with ridge regression as the first example. We then touch on the topic of nonparametric regression in a reproducing kernel Hilbert space (RKHS) via the kernel trick and kernel ridge regression. Some basic elements of the RKHS theory are presented, including the famous representer theorem. Lastly, we discuss the leave-one-out analysis and generalized cross-validation for tuning parameter selection in regularized linear models.

2.2 Multiple Linear Regression

Consider a *multiple linear regression* model:

$$Y = \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon, \quad (2.1)$$

where Y represents the *response* or *dependent variable* and the X variables are often called *explanatory variables* or *covariates* or *independent variables*. The intercept term can be included in the model by including 1 as one of the covariates, say $X_1 = 1$. Note that the term “random error” ε in (2.1) is a generic name used in statistics. In general, the “random error” here corresponds to the part of the response variable that cannot be explained or predicted by the covariates. It is often assumed that “random error” ε has zero mean, uncorrelated with covariates X , which is referred to as *exogenous* variables. Our goal is to estimate these β ’s, called *regression coefficients*, based on a random sample generated from model (2.1).

Suppose that $\{(X_{i1}, \dots, X_{ip}, Y_i)\}, i = 1, \dots, n$ is a random sample from

model (2.1). Then, we can write

$$Y_i = \sum_{j=1}^p X_{ij}\beta_j + \varepsilon_i. \quad (2.2)$$

The method of least-squares is a standard and popular technique for data fitting. It was advanced early in the nineteenth century by Gauss and Legendre. In (2.2) we have the residuals (r_i 's)

$$r_i = Y_i - \sum_{j=1}^p X_{ij}\beta_j.$$

Assume that random errors ε_i 's are *homoscedastic*, i.e., they are uncorrelated random variables with mean 0 and common variance σ^2 . The *least-squares method* is to minimize the residual sum-of-squares (RSS):

$$\text{RSS}(\boldsymbol{\beta}) = \sum_{i=1}^n r_i^2 = \sum_{i=1}^n (Y_i - \sum_{j=1}^p X_{ij}\beta_j)^2. \quad (2.3)$$

with respect to $\boldsymbol{\beta}$. Since (2.3) is a nice quadratic function of $\boldsymbol{\beta}$, there is a closed-form solution. Denote by

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \mathbf{X}_j = \begin{pmatrix} X_{1j} \\ \vdots \\ X_{nj} \end{pmatrix}, \mathbf{X} = \begin{pmatrix} X_{11} & \cdots & X_{1p} \\ \vdots & \cdots & \vdots \\ X_{n1} & \cdots & X_{np} \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Then (2.2) can be written in the matrix form

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

The matrix $\mathbf{X}$ is known as the *design matrix* and is of crucial importance to the whole theory of linear regression analysis. The $\text{RSS}(\boldsymbol{\beta})$ can be written as

$$\text{RSS}(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).$$

Differentiating $\text{RSS}(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ and setting the gradient vector to zero, we obtain the *normal equations*

$$\mathbf{X}^T \mathbf{Y} = \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}.$$

Here we assume that $p < n$ and $\mathbf{X}$ has rank p . Hence $\mathbf{X}^T \mathbf{X}$ is invertible and the normal equations yield the least-squares estimator of $\boldsymbol{\beta}$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}. \quad (2.4)$$

In this chapter $\mathbf{X}^T \mathbf{X}$ is assumed to be invertible unless specifically mentioned otherwise.

The fitted Y value is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y},$$

and the regression residual is

$$\hat{\mathbf{r}} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I} - \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T)\mathbf{Y}.$$

Theorem 2.1 Define $\mathbf{P} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$. Then we have

$$\mathbf{P}\mathbf{X}_j = \mathbf{X}_j, \quad j = 1, 2, \dots, p;$$

$$\mathbf{P}^2 = \mathbf{P} \quad \text{or} \quad \mathbf{P}(\mathbf{I}_n - \mathbf{P}) = \mathbf{0},$$

namely $\mathbf{P}$ is a projection matrix onto the space spanned by the columns of $\mathbf{X}$.

Proof. It follows from the direct calculation that

$$\mathbf{P}\mathbf{X} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X} = \mathbf{X}.$$

Taking the j column of the above equality, we obtain the first results. Similarly,

$$\mathbf{P}\mathbf{P} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = \mathbf{P}.$$

This completes the proof. ■

By Theorem 2.1 we can write

$$\hat{\mathbf{Y}} = \mathbf{P}\mathbf{Y}, \quad \hat{\mathbf{r}} = (\mathbf{I}_n - \mathbf{P})\mathbf{Y} \tag{2.5}$$

and we see two simple identities:

$$\mathbf{P}\hat{\mathbf{Y}} = \hat{\mathbf{Y}}, \quad \hat{\mathbf{Y}}^T\hat{\mathbf{r}} = 0.$$

This reveals an interesting geometric interpretation of the method of least-squares: the least-squares fit amounts to projecting the response vector onto the linear space spanned by the covariates. See Figure 2.1 for an illustration with two covariates.

2.2.1 The Gauss-Markov theorem

We assume the linear regression model (2.1) with

- *exogeneity*: $E(\varepsilon|X) = 0$;
- *homoscedasticity*: $\text{Var}(\varepsilon|X) = \sigma^2$.

Theorem 2.2 Under model (2.1) with exogenous and homoscedastic error, it follows that

- (i) (unbiasedness) $E(\hat{\boldsymbol{\beta}}|\mathbf{X}) = \boldsymbol{\beta}$.

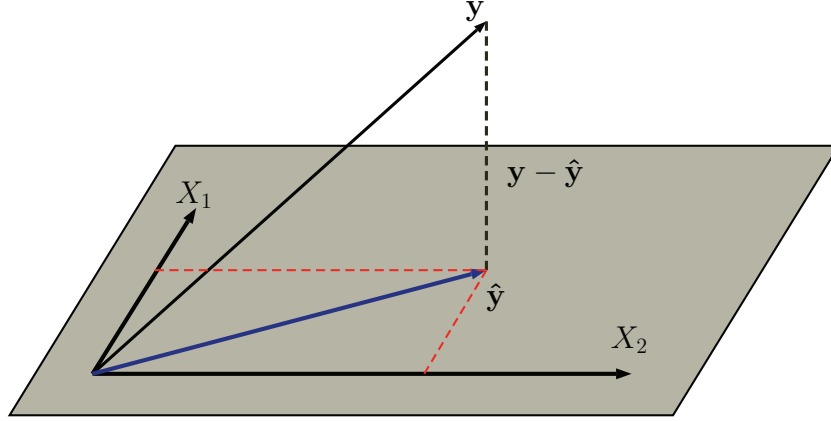


Figure 2.1: Geometric view of least-squares. The fitted value is the blue arrow, which is the projection of $\mathbf{Y}$ on the plane spanned by X_1 and X_2 .

- (ii) (conditional standard errors) $\text{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$.
- (iii) (BLUE) *The least-squares estimator $\hat{\beta}$ is the best linear unbiased estimator (BLUE). That is, for any given vector $\mathbf{a}$, $\mathbf{a}^T\hat{\beta}$ is a linear unbiased estimator of the parameter $\theta = \mathbf{a}^T\beta$. Further, for any linear unbiased estimator $\mathbf{b}^T\mathbf{Y}$ of θ , its variance is at least as large as that of $\mathbf{a}^T\hat{\beta}$.*

Proof. The first property follows directly from $E(\mathbf{Y}|\mathbf{X}) = \mathbf{X}\beta$ and

$$E(\hat{\beta}|\mathbf{X}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{X}\beta) = \beta.$$

To prove the second property, note that for any linear combination $\mathbf{A}\mathbf{Y}$, its variance-covariance matrix is given by

$$\text{Var}(\mathbf{A}\mathbf{Y}|\mathbf{X}) = \mathbf{A} \text{Var}(\mathbf{Y}|\mathbf{X})\mathbf{A}^T = \sigma^2\mathbf{A}\mathbf{A}^T. \quad (2.6)$$

Applying this formula to the least-squares estimator with $\mathbf{A} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$, we obtain the property (ii).

To prove property (iii), we first notice that $\mathbf{a}^T\hat{\beta}$ is an unbiased estimator of the parameter $\theta = \mathbf{a}^T\beta$, with the variance

$$\text{Var}(\mathbf{a}^T\hat{\beta}|\mathbf{X}) = \mathbf{a}^T \text{Var}(\hat{\beta}|\mathbf{X})\mathbf{a} = \sigma^2\mathbf{a}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{a}.$$

Now, consider any linear unbiased estimator, $\mathbf{b}^T\mathbf{Y}$, of the parameter θ . The unbiasedness requires that

$$\mathbf{b}^T\mathbf{X}\beta = \mathbf{a}^T\beta,$$

namely $\mathbf{X}^T \mathbf{b} = \mathbf{a}$. The variance of this linear estimator is

$$\sigma^2 \mathbf{b}^T \mathbf{b}.$$

To prove (iii), we need only to show that

$$\mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} \leq \mathbf{b}^T \mathbf{b}.$$

Note that

$$\mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} = \mathbf{b}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{b} = \mathbf{b}^T \mathbf{P} \mathbf{b}.$$

$\mathbf{P} = \mathbf{P}^2$ means that the eigenvalues of $\mathbf{P}$ are either 1 or 0 and hence $\mathbf{I}_n - \mathbf{P}$ is a semi-positive matrix. Thus,

$$\mathbf{b}^T (\mathbf{I}_n - \mathbf{P}) \mathbf{b} \geq 0,$$

or equivalently $\mathbf{b}^T \mathbf{b} \geq \mathbf{b}^T \mathbf{P} \mathbf{b}$. ■

Property (ii) of Theorem 2.2 gives the variance-covariance matrix of the least-squares estimate. In particular, the conditional standard error of $\hat{\beta}_i$ is simply $\sigma a_{ii}^{1/2}$ and the covariance between $\hat{\beta}_i$ and $\hat{\beta}_j$ is $\sigma^2 a_{ij}$, where a_{ij} is the (i, j) -th element of matrix $(\mathbf{X}^T \mathbf{X})^{-1}$.

In many applications σ^2 is often an unknown parameter of the model in addition to the regression coefficient vector $\boldsymbol{\beta}$. In order to use the variance-covariance formula, we first need to find a good estimate of σ^2 . Given the least-squares estimate of $\boldsymbol{\beta}$, RSS can be written as

$$\text{RSS} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = (\mathbf{Y} - \hat{\mathbf{Y}})^T (\mathbf{Y} - \hat{\mathbf{Y}}). \quad (2.7)$$

Define

$$\hat{\sigma}^2 = \text{RSS} / (n - p).$$

We will show in Theorem 2.3 that $\hat{\sigma}^2$ is an unbiased estimator of σ^2 .

Theorem 2.3 *Under the linear model (2.1) with homoscedastic error, it follows that*

$$\text{E}(\hat{\sigma}^2 | \mathbf{X}) = \sigma^2.$$

Proof. First by Theorem 2.1 we have

$$\text{RSS} = \|(\mathbf{I}_n - \mathbf{P})\mathbf{Y}\|^2 = \|(\mathbf{I}_n - \mathbf{P})(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\|^2 = \boldsymbol{\varepsilon}^T (\mathbf{I}_n - \mathbf{P}) \boldsymbol{\varepsilon}.$$

Let $\text{tr}(\mathbf{A})$ be the trace of the matrix $\mathbf{A}$. Using the property that $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$, we have

$$\text{RSS} = \text{tr}\{(\mathbf{I}_n - \mathbf{P})\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\}.$$

Hence,

$$\text{E}(\text{RSS} | \mathbf{X}) = \sigma^2 \text{tr}(\mathbf{I}_n - \mathbf{P}).$$

Because the eigenvalues of $\mathbf{P}$ are either 1 or 0, its trace is equal to its rank which is p under the assumption that $\mathbf{X}^T \mathbf{X}$ is invertible. Thus,

$$E(\hat{\sigma}^2 | \mathbf{X}) = \sigma^2(n - p)/(n - p) = \sigma^2.$$

This completes the proof. ■

2.2.2 Statistical tests

After fitting the regression model, we often need to perform some tests on the model parameters. For example, we may be interested in testing whether a particular regression coefficient should be zero, or whether several regression coefficients should be zero at the same time, which is equivalent to asking whether these variables are important in the presence of other covariates. To facilitate the discussion, we focus on the fixed design case where $\mathbf{X}$ is fixed. This is essentially the same as the random design case but conditioned upon the given realization $\mathbf{X}$.

We assume a homoscedastic model (2.1) with normal error. That is, ε is a Gaussian random variable with zero mean and variance σ^2 , written as $\varepsilon \sim N(0, \sigma^2)$. Note that

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\varepsilon}. \quad (2.8)$$

Then it is easy to see that

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2). \quad (2.9)$$

If we look at each $\hat{\beta}_j$ marginally, then $\hat{\beta}_j \sim N(\beta_j, v_j \sigma^2)$ where v_j is the j th diagonal element of $(\mathbf{X}^T \mathbf{X})^{-1}$. In addition,

$$(n - p) \hat{\sigma}^2 \sim \sigma^2 \chi_{n-p}^2 \quad (2.10)$$

and $\hat{\sigma}^2$ is independent of $\hat{\boldsymbol{\beta}}$. The latter can easily be shown as follow. By (2.7), $\hat{\sigma}^2$ depends on $\mathbf{Y}$ through $\mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I}_n - \mathbf{P})\boldsymbol{\varepsilon}$ whereas $\hat{\boldsymbol{\beta}}$ depends on $\mathbf{Y}$ through (2.8) or $\mathbf{X}^T \boldsymbol{\varepsilon}$. Note that both $(\mathbf{I}_n - \mathbf{P})\boldsymbol{\varepsilon}$ and $\mathbf{X}^T \boldsymbol{\varepsilon}$ are jointly normal because they are linear transforms of normally distributed random variables, and therefore their independence is equivalent to their uncorrelatedness. This can easily be checked by computing their covariance

$$E(\mathbf{I}_n - \mathbf{P})\boldsymbol{\varepsilon}(\mathbf{X}^T \boldsymbol{\varepsilon})^T = E(\mathbf{I}_n - \mathbf{P})\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^T \mathbf{X} = \sigma^2(\mathbf{I}_n - \mathbf{P})\mathbf{X} = \mathbf{0}.$$

If we want to test the hypothesis that $\beta_j = 0$, we can use the following t test statistic

$$t_j = \frac{\hat{\beta}_j}{\sqrt{v_j \hat{\sigma}^2}} \quad (2.11)$$

which follows a t -distribution with $n - p$ degrees of freedom under the null

hypothesis $H_0 : \beta_j = 0$. A level α test rejects the null hypothesis if $|t_j| > t_{n-p, 1-\alpha/2}$, where $t_{n-p, 1-\alpha/2}$ denotes the $100(1 - \alpha/2)$ percentile of the t -distribution with $n - p$ degrees of freedom.

In many applications the null hypothesis is that a subset of the covariates have zero regression coefficients. That is, this subset of covariates can be deleted from the regression model: they are unrelated to the response variable given the remaining variables. Under such a null hypothesis, we can reduce the model to a smaller model. Suppose that the reduced model has p_0 many regression coefficients. Let RSS and RSS_0 be the residual sum-of-squares based on the least-squares fit of the full model and the reduced smaller model, respectively. If the null hypothesis is true, then these two quantities should be similar: The RSS reduction by using the full model is small, in relative terms. This leads to the *F-statistic*:

$$F = \frac{(\text{RSS}_0 - \text{RSS})/(p - p_0)}{\text{RSS}/(n - p)}. \quad (2.12)$$

Under the null hypothesis that the reduced model is correct, $F \sim F_{p-p_0, n-p}$.

The normal error assumption can be relaxed if the sample size n is large. First, we know that $(\mathbf{X}^T \mathbf{X})^{\frac{1}{2}}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma$ always has zero mean and an identity variance-covariance matrix. On the other hand, (2.8) gives us

$$(\mathbf{X}^T \mathbf{X})^{\frac{1}{2}}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma = (\mathbf{X}^T \mathbf{X})^{-\frac{1}{2}} \mathbf{X}^T \boldsymbol{\varepsilon}/\sigma.$$

Observe that $(\mathbf{X}^T \mathbf{X})^{-\frac{1}{2}} \mathbf{X}^T \boldsymbol{\varepsilon}/\sigma$ is a linear combination of n i.i.d. random variables $\{\varepsilon_i\}_{i=1}^n$ with zero mean and variance 1. Then the central limit theorem implies that under some regularity conditions,

$$\hat{\boldsymbol{\beta}} \xrightarrow{D} N(\boldsymbol{\beta}, (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2). \quad (2.13)$$

Consequently, when n is large, the distribution of the t test statistic in (2.11) is approximately $N(0, 1)$, and the distribution of the F test statistic in (2.12) is approximately $\chi_{p-p_0}^2/(p - p_0)$.

2.3 Weighted Least-Squares

The method of least-squares can be further generalized to handle the situations where errors are *heteroscedastic* or correlated. In the linear regression model (2.2), we would like to keep the assumption $E(\varepsilon|\mathbf{X}) = 0$ which means there is no structure information left in the error term. However, the constant variance assumption $\text{Var}(\varepsilon_i|\mathbf{X}_i) = \sigma^2$ may not likely hold in many applications. For example, if y_i is the average response value of the i th subject in a study in which k_i many repeated measurements have been taken, then it would be more reasonable to assume $\text{Var}(\varepsilon_i|\mathbf{X}_i) = \sigma^2/k_i$.

Let us consider a modification of model (2.1) as follows

$$Y_i = \sum_{j=1}^p X_{ij} \beta_j + \varepsilon_i; \quad \text{Var}(\varepsilon_i|\mathbf{X}_i) = \sigma^2 v_i \quad (2.14)$$

where v_i s are known positive constants but σ^2 remains unknown. One can still use the ordinary least-squares (OLS) estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$. It is easy to show that the OLS estimator is unbiased but no longer BLUE. In fact, the OLS estimator can be improved by using the *weighted least-squares* method.

Let $Y_i^* = v_i^{-1/2} Y_i$, $X_{ij}^* = v_i^{-1/2} X_{ij}$, $\varepsilon_i^* = v_i^{-1/2} \varepsilon_i$. Then the new model (2.14) can be written as

$$Y_i^* = \sum_{j=1}^p X_{ij}^* \beta_j + \varepsilon_i^* \quad (2.15)$$

with $\text{Var}(\varepsilon_i^* | \mathbf{X}_i^*) = \sigma^2$. Therefore, the working data $\{(X_{i1}^*, \dots, X_{ip}^*, Y_i^*)\}_{i=1}^n$ obey the standard *homoscedastic* linear regression model. Applying the standard least-squares method to the working data, we have

$$\hat{\boldsymbol{\beta}}^{wls} = \underset{\boldsymbol{\beta}}{\text{argmin}} \sum_{i=1}^n \left(Y_i^* - \sum_{j=1}^p X_{ij}^* \beta_j \right)^2 = \underset{\boldsymbol{\beta}}{\text{argmin}} \sum_{i=1}^n v_i^{-1} \left(Y_i - \sum_{j=1}^p X_{ij} \beta_j \right)^2.$$

It follows easily from Theorem 2.2 that the weighted least-squares estimator is the BLUE for $\boldsymbol{\beta}$.

In model (2.14) the errors are assumed to be uncorrelated. In general, the method of least-squares can be extended to handle heteroscedastic and correlated errors.

Assume that

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

and the variance-covariance matrix of $\boldsymbol{\varepsilon}$ is given

$$\text{Var}(\boldsymbol{\varepsilon} | \mathbf{X}) = \sigma^2 \mathbf{W}, \quad (2.16)$$

in which $\mathbf{W}$ is a known positive definite matrix. Let $\mathbf{W}^{-1/2}$ be the square root of $\mathbf{W}^{-1}$, i.e.,

$$(\mathbf{W}^{-1/2})^T \mathbf{W}^{-1/2} = \mathbf{W}^{-1}.$$

Then

$$\text{Var}(\mathbf{W}^{-1/2} \boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I},$$

which are homoscedastic and uncorrelated.

Define the working data as follows:

$$\mathbf{Y}^* = \mathbf{W}^{-1/2} \mathbf{Y}, \quad \mathbf{X}^* = \mathbf{W}^{-1/2} \mathbf{X}, \quad \boldsymbol{\varepsilon}^* = \mathbf{W}^{-1/2} \boldsymbol{\varepsilon}.$$

Then we have

$$\mathbf{Y}^* = \mathbf{X}^* \boldsymbol{\beta} + \boldsymbol{\varepsilon}^*. \quad (2.17)$$

Thus, we can apply the standard least-squares to the working data. First, the residual sum-of-squares (RSS) is

$$\text{RSS}(\boldsymbol{\beta}) = \|\mathbf{Y}^* - \mathbf{X}^* \boldsymbol{\beta}\|^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T \mathbf{W}^{-1} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}). \quad (2.18)$$

Then the *general least-squares* estimator is defined by

$$\begin{aligned}\hat{\beta} &= \operatorname{argmin}_{\beta} \operatorname{RSS}(\beta) \\ &= (\mathbf{X}^{*T} \mathbf{X}^*)^{-1} \mathbf{X}^{*T} \mathbf{Y}^* \\ &= (\mathbf{X}^T \mathbf{W}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{-1} \mathbf{Y}.\end{aligned}\tag{2.19}$$

Again, $\hat{\beta}$ is the BLUE according to Theorem 2.2.

In practice, it is difficult to know precisely the $n \times n$ covariance matrix $\mathbf{W}$; the misspecification of $\mathbf{W}$ in the general least-squares seems hard to avoid. Let us examine the robustness of the general least-squares estimate. Assume that $\operatorname{Var}(\varepsilon) = \sigma^2 \mathbf{W}_0$, where $\mathbf{W}_0$ is unknown to us, but we employ the general least-squares method (2.19) with the wrong covariance matrix $\mathbf{W}$. We can see that the general least-squares estimator is still unbiased:

$$\mathbb{E}(\hat{\beta}|\mathbf{X}) = (\mathbf{X}^T \mathbf{W}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{-1} \mathbf{X} \beta = \beta.$$

Furthermore, the variance-covariance matrix is given by

$$\operatorname{Var}(\hat{\beta}) = (\mathbf{X}^T \mathbf{W}^{-1} \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{W}^{-1} \mathbf{W}_0 \mathbf{W}^{-1} \mathbf{X}) (\mathbf{X}^T \mathbf{W}^{-1} \mathbf{X})^{-1},$$

which is of order $O(n^{-1})$ under some mild conditions. In other words, using the wrong covariance matrix would still give us a root- n consistent estimate. So even when errors are heteroscedastic and correlated, the ordinary least-squares estimate with $\mathbf{W} = \mathbf{I}$ and the weighted least-squares estimate with $\mathbf{W} = \operatorname{diag}(\mathbf{W}_0)$ still give us an unbiased and $n^{-1/2}$ consistent estimator. Of course, we still prefer using a working $\mathbf{W}$ matrix that is identical or close to the true $\mathbf{W}_0$.

2.4 Box-Cox Transformation

In practice we often take a transformation of the response variable before fitting a linear regression model. The idea is that the transformed response variable can be modeled by the set of covariates via the classical multiple linear regression model. For example, in many engineering problems we expect $Y \propto X_1^{\beta_1} X_2^{\beta_2} \cdots X_p^{\beta_p}$ where all variables are positive. Then a linear model seems proper by taking logarithms: $\log(Y) = \sum_{j=1}^p \beta_j X_j + \varepsilon$. If we assume $\varepsilon \sim N(0, \sigma^2)$, then in the original scale the model is $Y = (\prod_j X_j^{\beta_j}) \varepsilon^*$ where ε^* is a log-normal random variable: $\log \varepsilon^* \sim N(0, \sigma^2)$.

Box and Cox (1964) advocated the variable transformation idea in linear regression and also proposed a systematic way to estimate the transformation function from data. Their method is now known as the *Box-Cox transform* in the literature. Box and Cox (1964) suggested a parametric family for the transformation function. Let $Y^{(\lambda)}$ denote the transformed response where λ parameterizes the transformation function:

$$Y^{(\lambda)} = \begin{cases} \frac{Y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(Y) & \text{if } \lambda = 0 \end{cases}.$$

The Box-Cox model assumes that

$$Y^{(\lambda)} = \sum_{j=1}^p X_j \beta_j + \varepsilon$$

where $\varepsilon \sim N(0, \sigma^2)$.

The likelihood function of the Box-Cox model is given by

$$L(\lambda, \boldsymbol{\beta}, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma}}\right)^n e^{-\frac{1}{2\sigma^2} \|\mathbf{Y}^{(\lambda)} - \mathbf{X}\boldsymbol{\beta}\|^2} \cdot J(\lambda, \mathbf{Y})$$

where $J(\lambda, \mathbf{Y}) = \prod_{i=1}^n \left| \frac{dy_i^{(\lambda)}}{dy_i} \right| = \left(\prod_{i=1}^n |y_i| \right)^{\lambda-1}$. Given λ , the maximum likelihood estimators (MLE) of $\boldsymbol{\beta}$ and σ^2 are obtained by the ordinary least-squares:

$$\hat{\boldsymbol{\beta}}(\lambda) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}^{(\lambda)}, \quad \hat{\sigma}^2(\lambda) = \frac{1}{n} \|\mathbf{Y}^{(\lambda)} - \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}^{(\lambda)}\|^2.$$

Plugging $\hat{\boldsymbol{\beta}}(\lambda), \hat{\sigma}^2(\lambda)$ into $L(\lambda, \boldsymbol{\beta}, \sigma^2)$ yields a likelihood function of λ

$$\log L(\lambda) = (\lambda - 1) \sum_{i=1}^n \log(|y_i|) - \frac{n}{2} \log \hat{\sigma}^2(\lambda) - \frac{n}{2}.$$

Then the MLE of λ is

$$\hat{\lambda}_{mle} = \operatorname{argmax}_{\lambda} \log L(\lambda),$$

and the MLE of $\boldsymbol{\beta}$ and σ^2 are $\hat{\boldsymbol{\beta}}(\hat{\lambda}_{mle})$ and $\hat{\sigma}^2(\hat{\lambda}_{mle})$, respectively.

2.5 Model Building and Basis Expansions

Multiple linear regression can be used to produce nonlinear regression and other very complicated models. The key idea is to create new covariates from the original ones by adopting some transformations. We then fit a multiple linear regression model using augmented covariates.

For simplicity, we first illustrate some useful transformations in the case of $p = 1$, which is closely related to the curve fitting problem in *nonparametric regression*. In a nonparametric regression model

$$Y = f(X) + \varepsilon,$$

we do not assume a specific form of the regression function $f(x)$, but assume only some qualitative aspects of the regression function. Examples include that $f(\cdot)$ is continuous with a certain number of derivatives or that $f(\cdot)$ is convex. The aim is to estimate the function $f(x)$ and its derivatives, without a specific parametric form of $f(\cdot)$. See, for example Fan and Gijbels (1996), Li and Racine (2007), Hastie, Tibshirani and Friedman (2009), among others.

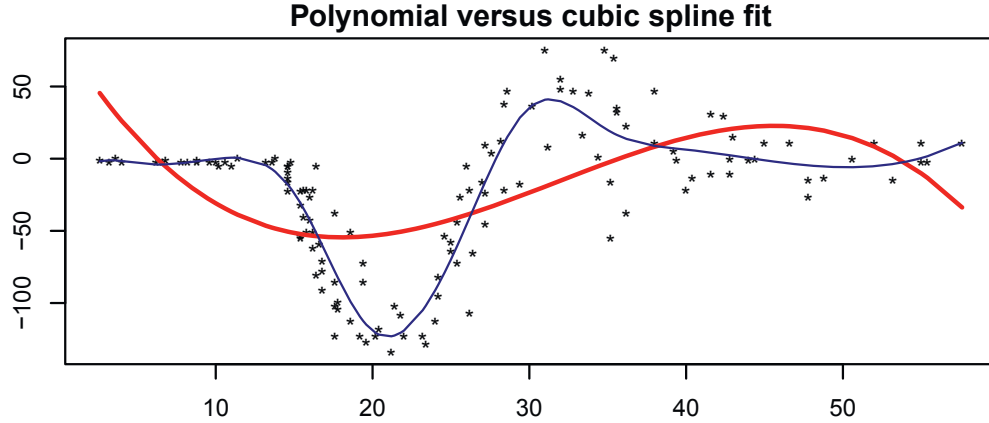


Figure 2.2: Scatter plot of time (in milliseconds) after a simulated impact on motorcycles against the head acceleration of a test object. Red = cubic polynomial fit, blue = cubic spline fit.

2.5.1 Polynomial regression

Without loss of generality, assume X is bounded on $[0, 1]$ for simplicity. The Weierstrass approximation theorem states that any continuous $f(x)$ can be uniformly approximated by a polynomial function up to any precision factor. Let us approximate the model by

$$Y = \underbrace{\beta_0 + \beta_1 X + \cdots + \beta_d X^d}_{\approx f(X)} + \varepsilon$$

This *polynomial regression* is a multiple regression problem by setting $X_0 = 1, X_1 = X, \dots, X_d = X^d$. The design matrix now becomes

$$\mathbf{B}_1 = \begin{pmatrix} 1 & x_1 & \cdots & x_1^d \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_n & \cdots & x_n^d \end{pmatrix}.$$

We estimate $f(x)$ by

$$\hat{f}(x) = \hat{\beta}_0 + \sum_{m=1}^d \hat{\beta}_m x^m,$$

where $\hat{\beta} = (\mathbf{B}_1^T \mathbf{B}_1)^{-1} \mathbf{B}_1^T \mathbf{Y}$ is the least-squares estimate.

Polynomial functions have derivatives everywhere and are global functions. They are not very flexible in approximating functions with local features such as functions with various degrees of smoothness at different locations. Figure 2.2 shows the cubic polynomial fit to a motorcycle data. Clearly, it does not fit the data very well. Increasing the order of the polynomial fits will

help reduce the bias issue, but will not solve the lack of fit issue. This is because the underlying function cannot be economically approximated by a polynomial function. It requires high-order polynomials to reduce approximation biases, but this increases both variances and instability of the fits. This leads to the introduction of spline functions that allow for more flexibility in function approximation.

2.5.2 Spline regression

Let $\tau_0 < \tau_1 < \dots < \tau_{K+1}$. A *spline function* of degree d on $[\tau_0, \tau_{K+1}]$ is a piecewise polynomial function of degree d on intervals $[\tau_j, \tau_{j+1})$ ($j = 0, \dots, K$), with continuous first $d-1$ derivatives. The points where the spline function might not have continuous d^{th} derivatives are $\{\tau_j\}_{j=1}^K$, which are called *knots*. Thus, a cubic spline function is a piecewise polynomial function with continuous first two derivatives and the points where the third derivative might not exist are called knots of the cubic spline. An example of a cubic fit is given by Figure 2.2.

All spline functions of degree d form a linear space. Let us determine its basis functions.

Linear Splines: A continuous function on $[0, 1]$ can also be approximated by a piecewise constant or linear function. We wish to use a continuous function to approximate $f(x)$. Since a piecewise constant function is not continuous unless the function is a constant in the entire interval, we use a continuous piecewise linear function to fit $f(x)$. Suppose that we split the interval $[0, 1]$ into three regions: $[0, \tau_1]$, $[\tau_1, \tau_2]$, $[\tau_2, 1]$ with given knots τ_1, τ_2 . Denote by $l(x)$ the continuous piecewise linear function. In the first interval $[0, \tau_1]$ we write

$$l(x) = \beta_0 + \beta_1 x, \quad x \in [0, \tau_1],$$

as it is linear. Since $l(x)$ must be continuous at τ_1 , the newly added linear function must have an intercept 0 at point τ_1 . Thus, in $[\tau_1, \tau_2]$ we must have

$$l(x) = \beta_0 + \beta_1 x + \beta_2(x - \tau_1)_+, \quad x \in [\tau_1, \tau_2],$$

where z_+ equals z if $z > 0$ and zero otherwise. The function is linear in $[\tau_1, \tau_2]$ with slope $\beta_1 + \beta_2$. Likewise, in $[\tau_2, 1]$ we write

$$l(x) = \beta_0 + \beta_1 x + \beta_2(x - \tau_1)_+ + \beta_3(x - \tau_2)_+, \quad x \in [\tau_2, 1].$$

The function is now clearly a piecewise linear function with possible different slopes on different intervals. Therefore, the basis functions are

$$B_0(x) = 1, B_1(x) = x, B_2(x) = (x - \tau_1)_+, B_3(x) = (x - \tau_2)_+; \quad (2.20)$$

which are called a *linear spline* basis. We then approximate the nonparametric regression model as

$$Y = \underbrace{\beta_0 B_0(X) + \beta_1 B_1(X) + \beta_2 B_2(X) + \beta_3 B_3(X)}_{\approx f(X)} + \varepsilon.$$

This is again a multiple regression problem where we set $X_0 = B_0(X)$, $X_1 = B_1(X)$, $X_2 = B_2(X)$, $X_3 = B_3(X)$. The corresponding design matrix becomes

$$\mathbf{B}_2 = \begin{pmatrix} 1 & x_1 & (x_1 - \tau_1)_+ & (x_1 - \tau_2)_+ \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & (x_n - \tau_1)_+ & (x_n - \tau_2)_+ \end{pmatrix},$$

and we estimate $f(x)$ by

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 (x - \tau_1)_+ + \hat{\beta}_3 (x - \tau_2)_+,$$

where $\hat{\beta} = (\mathbf{B}_2^T \mathbf{B}_2)^{-1} \mathbf{B}_2^T \mathbf{Y}$. The above method applies more generally to a multiple knot setting for the data on any intervals.

Cubic Splines: We can further consider fitting piecewise polynomials whose derivatives are also continuous. A popular choice is the so-called cubic spline that is a piecewise cubic polynomial function with continuous first and second derivatives. Again, we consider two knots and three regions: $[0, \tau_1]$, $[\tau_1, \tau_2]$, $[\tau_2, 1]$. Let $c(x)$ be a cubic spline. In $[0, \tau_1]$ we write

$$c(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3, \quad x \leq \tau_1.$$

And $c(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \delta(x)$ in $[\tau_1, \tau_2]$. By definition, $\delta(x)$ is a cubic function in $[\tau_1, \tau_2]$ and its first and second derivatives equal zero at $x = \tau_1$. Then we must have

$$\delta(x) = \beta_4 (x - \tau_1)_+^3, \quad x \in [\tau_1, \tau_2]$$

which means

$$c(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \tau_1)_+^3, \quad x \in [\tau_1, \tau_2].$$

Likewise, in $[\tau_2, 1]$ we must have

$$c(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \tau_1)_+^3 + \beta_5 (x - \tau_2)_+^3, \quad x > \tau_2.$$

Therefore, the basis functions are

$$\begin{aligned} B_0(x) &= 1, B_1(x) = x, B_2(x) = x^2, B_3(x) = x^3 \\ B_4(x) &= (x - \tau_1)_+^3, B_5(x) = (x - \tau_2)_+^3. \end{aligned}$$

The corresponding transformed design matrix becomes

$$\mathbf{B}_3 = \begin{pmatrix} 1 & x_1 & x_1^2 & x_1^3 & (x_1 - \tau_1)_+^3 & (x_1 - \tau_2)_+^3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 & (x_n - \tau_1)_+^3 & (x_n - \tau_2)_+^3 \end{pmatrix},$$

and we estimate $f(x)$ by

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 + \hat{\beta}_3 x^3 + \hat{\beta}_4 (x - \tau_1)_+^3 + \hat{\beta}_5 (x - \tau_2)_+^3,$$

where $\hat{\beta} = (\mathbf{B}_3^T \mathbf{B}_3)^{-1} \mathbf{B}_3^T \mathbf{Y}$ is the least-squares estimate of the coefficients.

In general, if there are K knots $\{\tau_1, \dots, \tau_K\}$, then the *basis functions* of *cubic splines* are

$$\begin{aligned} B_0(x) &= 1, B_1(x) = x, B_2(x) = x^2, B_3(x) = x^3 \\ B_4(x) &= (x - \tau_1)_+^3, \dots, B_{K+3}(x) = (x - \tau_K)_+^3. \end{aligned}$$

By approximating the nonparametric function $f(X)$ by the spline function with knots $\{\tau_j\}_{j=1}^K$, we have

$$Y = \underbrace{\beta_0 B_0(X) + \beta_1 B_1(X) + \dots + \beta_{K+3} B_{K+3}(X)}_{\approx f(X)} + \varepsilon \quad (2.21)$$

This *spline regression* is again a multiple regression problem.

Natural Cubic Splines: Extrapolation is always a serious issue in regression. It is not wise to fit a cubic function to a region where the observations are scarce. If we must, extrapolation with a linear function is preferred. A *natural cubic spline* is a special cubic spline with additional constraints: the cubic spline must be linear beyond two end knots. Consider a natural cubic spline, $\text{NC}(x)$, with knots at $\{\tau_1, \dots, \tau_K\}$. By its cubic spline representation, we can write

$$\text{NC}(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \sum_{j=1}^K \beta_{3+j} (x - \tau_j)_+^3.$$

First, $\text{NC}(x)$ is linear for $x < \tau_1$, which implies that

$$\beta_2 = \beta_3 = 0.$$

Second, $\text{NC}(x)$ is linear for $x > \tau_K$, which means that

$$\sum_{j=1}^K \beta_{3+j} = 0, \quad \sum_{j=1}^K \tau_j \beta_{3+j} = 0,$$

corresponding to the coefficients for the cubic and quadratic term of the polynomial $\sum_{j=1}^K \beta_{3+j} (x - \tau_j)^3$ for $x > \tau_K$. We solve for β_{K+2}, β_{K+3} from the above equations and then write $\text{NC}(x)$ as

$$\text{NC}(x) = \sum_{j=0}^{K-1} \beta_j B_j(x),$$

where the *natural cubic spline* basis functions are given by

$$\begin{aligned} B_0(x) &= 1, B_1(x) = x, \\ B_{j+1}(x) &= \frac{(x - \tau_j)_+^3 - (x - \tau_K)_+^3}{\tau_j - \tau_K} - \frac{(x - \tau_{K-1})_+^3 - (x - \tau_K)_+^3}{\tau_{K-1} - \tau_K} \\ &\text{for } j = 1, \dots, K - 2. \end{aligned}$$

Again, by approximating the nonparametric function with the natural cubic spline, we have

$$Y = \sum_{j=0}^{K-1} \beta_j B_j(X) + \varepsilon. \quad (2.22)$$

which can be solved by using multiple regression techniques.

2.5.3 Multiple covariates

The concept of polynomial regression extends to multivariate covariates. The simplest example is the bivariate regression model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_1 X_2 + \beta_5 X_2^2 + \varepsilon.$$

The term $X_1 X_2$ is called the *interaction*, which quantifies how X_1 and X_2 work together to contribute to the response. Often, one introduces interactions without using the quadratic term, leading to a slightly simplified model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon.$$

More generally, the multivariate quadratic regression is of the form

$$Y = \sum_{j=1}^p \beta_j X_j + \sum_{j < k} \beta_{jk} X_j X_k + \varepsilon \quad (2.23)$$

and the multivariate regression with main effects (the linear terms) and interactions is of the form

$$Y = \sum_{j=1}^p \beta_j X_j + \sum_{j < k} \beta_{jk} X_j X_k + \varepsilon. \quad (2.24)$$

This concept can also be extended to the multivariate spline case. The basis function can be the tensor of the univariate spline basis function for not only unstructured $f(\mathbf{x})$, but also other basis functions for structured $f(\mathbf{x})$. Unstructured nonparametric functions are not very useful: If each variable uses 100 basis functions, then there are 100^p basis functions in the tensor products, which is prohibitively large for say, $p = 10$. Such an issue is termed the “curse-of-dimensionality” in literature. See Hastie and Tibshirani (1990) and Fan and Gijbels (1996). On the other hand, for the structured multivariate

model, such as the following additive model (Stone, 1985, 1994; Hastie and Tibshirani, 1990),

$$Y = f_1(X_1) + \cdots + f_p(X_p) + \varepsilon \quad (2.25)$$

the basis functions are simply the collection of all univariate basis functions for approximating $f_1, \dots, f_p$. The total number grows only linearly with p .

In general, let $B_m(\mathbf{x})$ be the basis functions $m = 1, \dots, M$. Then, we approximate the multivariate nonparametric regression model $Y = f(\mathbf{X}) + \varepsilon$ by

$$Y = \sum_{m=1}^M \beta_j B_j(\mathbf{X}) + \varepsilon. \quad (2.26)$$

This can be fit using a multiple regression technique. The new design matrix is

$$\mathbf{B} = \begin{pmatrix} B_1(\mathbf{X}_1) & \cdots & B_M(\mathbf{X}_1) \\ \vdots & \cdots & \vdots \\ B_1(\mathbf{X}_n) & \cdots & B_M(\mathbf{X}_n) \end{pmatrix}$$

and the least-squares estimate is given by

$$\hat{f}(\mathbf{x}) = \sum_{m=1}^M \hat{\beta}_m B_m(\mathbf{x}),$$

where

$$\hat{\beta} = (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T \mathbf{Y}.$$

The above fitting implicitly assumes that $M \ll n$. This condition in fact can easily be violated in unstructured multivariate nonparametric regression. For the additive model (2.25), in which we assume $f(\mathbf{x}) = \sum_{j=1}^p f_j(x_j)$ where each $f_j(x_j)$ is a smooth univariate function of x_j , the univariate basis expansion ideas can be readily applied to approximation of each $f_j(x_j)$:

$$f_j(x_j) \approx \sum_{m=1}^{M_j} B_{jm}(x_j) \beta_{jm}$$

which implies that the fitted regression function is

$$f(\mathbf{x}) \approx \sum_{j=1}^p \sum_{m=1}^{M_j} B_{jm}(x_j) \beta_{jm}.$$

In Section 2.6.5 and Section 2.7 we introduce a fully nonparametric multiple regression technique which can be regarded as a basis expansion method where the basis functions are given by kernel functions.

2.6 Ridge Regression

2.6.1 Bias-variance tradeoff

Recall that the ordinary least squares estimate is defined by $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$ when $\mathbf{X}$ is of full rank. In practice, we often encounter highly correlated covariates, which is known as the *collinearity* issue. As a result, although $\mathbf{X}^T \mathbf{X}$ is still invertible, its smallest eigenvalue can be very small. Under the homoscedastic error model, the variance-covariance matrix of the OLS estimate is $\text{Var}(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$. Thus, the collinearity issue makes $\text{Var}(\hat{\beta})$ large.

Hoerl and Kennard (1970) introduced the *ridge regression* estimator as follows:

$$\hat{\beta}_\lambda = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{Y}, \quad (2.27)$$

where $\lambda > 0$ is a regularization parameter. In the usual case ($\mathbf{X}^T \mathbf{X}$ is invertible), ridge regression reduces to OLS by setting $\lambda = 0$. However, ridge regression is always well defined even when $\mathbf{X}$ is not full rank.

Under the assumption $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}$, it is easy to show that

$$\text{Var}(\hat{\beta}_\lambda) = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \sigma^2. \quad (2.28)$$

We always have $\text{Var}(\hat{\beta}_\lambda) < (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$. The ridge regression estimator reduces the estimation variance by paying a price in estimation bias:

$$\text{E}(\hat{\beta}_\lambda) - \beta = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{X} \beta - \beta = -\lambda (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \beta. \quad (2.29)$$

The overall estimation accuracy is gauged by the mean squared error (MSE). For $\hat{\beta}_\lambda$ its MSE is given by

$$\text{MSE}(\hat{\beta}_\lambda) = \text{E}(\|\hat{\beta}_\lambda - \beta\|^2). \quad (2.30)$$

By (2.28) and (2.29) we have

$$\begin{aligned} \text{MSE}(\hat{\beta}_\lambda) &= \text{tr}((\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \sigma^2) \\ &\quad + \lambda^2 \beta^T (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-2} \beta \\ &= \text{tr}\left((\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-2} [\lambda^2 \beta \beta^T + \sigma^2 \mathbf{X}^T \mathbf{X}]\right). \end{aligned} \quad (2.31)$$

It can be shown that $\frac{d\text{MSE}(\hat{\beta}_\lambda)}{d\lambda}|_{\lambda=0} < 0$, which implies that there are some proper λ values by which ridge regression improves OLS.

2.6.2 ℓ_2 penalized least squares

Define a penalized residual sum-of-squares (PRSS) as follows:

$$\text{PRSS}(\boldsymbol{\beta}|\lambda) = \sum_{i=1}^n (Y_i - \sum_{j=1}^p X_{ij}\beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2. \quad (2.32)$$

Then let

$$\hat{\boldsymbol{\beta}}_\lambda = \underset{\boldsymbol{\beta}}{\text{argmin}} \text{PRSS}(\boldsymbol{\beta}|\lambda). \quad (2.33)$$

Note that we can write it in a matrix form

$$\text{PRSS}(\boldsymbol{\beta}|\lambda) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2.$$

The term $\lambda\|\boldsymbol{\beta}\|^2$ is called the ℓ_2 -penalty of $\boldsymbol{\beta}$. Taking derivatives with respect to $\boldsymbol{\beta}$ and setting it to zero, we solve the root of the following equation

$$-\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) + \lambda\boldsymbol{\beta} = 0,$$

which yields

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{Y}.$$

The above discussion shows that ridge regression is equivalent to the ℓ_2 penalized least-squares.

We have seen that ridge regression can achieve a smaller MSE than OLS. In other words, the ℓ_2 penalty term helps regularize (reduce) estimation variance and produces a better estimator when the reduction in variance exceeds the induced extra bias. From this perspective, one can also consider a more general ℓ_q penalized least-squares estimate

$$\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_{j=1}^p |\beta_j|^q \quad (2.34)$$

where q is a positive constant. This is referred to as the **Bridge estimator** (Frank and Friedman, 1993). The ℓ_q penalty is strictly concave when $0 < q < 1$, and strictly convex when $q > 1$. For $q = 1$, the resulting ℓ_1 penalized least-squares is also known as the Lasso (Tibshirani, 1996). Chapter 3 covers the Lasso in great detail. Among all Bridge estimators only the ridge regression has a nice closed-form solution with a general design matrix.

2.6.3 Bayesian interpretation

Ridge regression has a neat Bayesian interpretation in the sense that it can be a formal Bayes estimator. We begin with the homoscedastic Gaussian error model:

$$Y_i = \sum_{j=1}^p X_{ij}\beta_j + \varepsilon_i$$

and $\varepsilon_i | \mathbf{X}_i \sim N(0, \sigma^2)$. Now suppose that β_j 's are also independent $N(0, \tau^2)$ variables, which represent our knowledge about the regression coefficients before seeing the data. In Bayesian statistics, $N(0, \tau^2)$ is called the prior distribution of β_j . The model and the prior together give us the posterior distribution of $\boldsymbol{\beta}$ given the data (the conditional distribution of $\boldsymbol{\beta}$ given $\mathbf{Y}, \mathbf{X}$). Straightforward calculations yield

$$P(\boldsymbol{\beta} | \mathbf{Y}, \mathbf{X}) \propto \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2\right) \exp\left(-\frac{1}{2\tau^2} \|\boldsymbol{\beta}\|^2\right). \quad (2.35)$$

A maximum posteriori probability (MAP) estimate is defined as

$$\begin{aligned} \hat{\boldsymbol{\beta}}^{\text{MAP}} &= \operatorname{argmax}_{\boldsymbol{\beta}} P(\boldsymbol{\beta} | \mathbf{Y}, \mathbf{X}) \\ &= \operatorname{argmax}_{\boldsymbol{\beta}} \left\{ -\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 - \frac{1}{2\tau^2} \|\boldsymbol{\beta}\|^2 \right\}. \end{aligned} \quad (2.36)$$

It is easy to see that $\hat{\boldsymbol{\beta}}^{\text{MAP}}$ is ridge regression with $\lambda = \frac{\sigma^2}{\tau^2}$. Another popular Bayesian estimate is the posterior mean. In this model, the posterior mean and posterior mode are the same.

From the Bayesian perspective, it is easy to construct a generalized ridge regression estimator. Suppose that the prior distribution for the entire $\boldsymbol{\beta}$ vector is $N(0, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ is a general positive definite matrix. Then the posterior distribution is computed as

$$P(\boldsymbol{\beta} | \mathbf{Y}, \mathbf{X}) \propto \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2\right) \exp\left(-\frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\beta}\right). \quad (2.37)$$

The corresponding MAP estimate is

$$\begin{aligned} \hat{\boldsymbol{\beta}}^{\text{MAP}} &= \operatorname{argmax}_{\boldsymbol{\beta}} P(\boldsymbol{\beta} | \mathbf{Y}, \mathbf{X}) \\ &= \operatorname{argmax}_{\boldsymbol{\beta}} \left\{ -\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 - \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\beta} \right\}. \end{aligned} \quad (2.38)$$

It is easy to see that

$$\hat{\boldsymbol{\beta}}^{\text{MAP}} = (\mathbf{X}^T \mathbf{X} + \sigma^2 \boldsymbol{\Sigma}^{-1})^{-1} \mathbf{X}^T \mathbf{Y}. \quad (2.39)$$

This generalized ridge regression can take into account different scales of covariates, by an appropriate choice of $\boldsymbol{\Sigma}$.

2.6.4 Ridge regression solution path

The performance of ridge regression heavily depends on the choice of λ . In practice we only need to compute ridge regression estimates at a fine grid of λ values and then select the best from these candidate solutions. Although ridge regression is easy to compute for a λ owing to its nice closed-form solution expression, the total cost could be high if the process is repeated many times.

Through a more careful analysis, one can see that the solutions of ridge regression at a fine grid of λ values can be computed very efficiently via singular value decomposition.

Assume $n > p$ and $\mathbf{X}$ is full rank. The singular value decomposition (SVD) of $\mathbf{X}$ is given by

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

where $\mathbf{U}$ is a $n \times p$ orthogonal matrix, $\mathbf{V}$ is a $p \times p$ orthogonal matrix and $\mathbf{D}$ is a $p \times p$ diagonal matrix whose diagonal elements are the ordered (from large to small) singular values of $\mathbf{X}$. Then

$$\mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{D}\mathbf{U}^T\mathbf{U}\mathbf{D}\mathbf{V}^T = \mathbf{V}\mathbf{D}^2\mathbf{V}^T,$$

$$\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I} = \mathbf{V}\mathbf{D}^2\mathbf{V}^T + \lambda\mathbf{I} = \mathbf{V}(\mathbf{D}^2 + \lambda\mathbf{I})\mathbf{V}^T,$$

$$(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1} = \mathbf{V}(\mathbf{D}^2 + \lambda\mathbf{I})^{-1}\mathbf{V}^T.$$

The ridge regression estimator $\hat{\boldsymbol{\beta}}_\lambda$ can now be written as

$$\begin{aligned} \hat{\boldsymbol{\beta}}_\lambda &= (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{Y} \\ &= \mathbf{V}(\mathbf{D}^2 + \lambda\mathbf{I})^{-1}\mathbf{D}\mathbf{U}^T\mathbf{Y} \\ &= \sum_{j=1}^p \frac{d_j}{d_j^2 + \lambda} \langle \mathbf{U}_j, \mathbf{Y} \rangle \mathbf{V}_j, \end{aligned} \tag{2.40}$$

where d_j is the j^{th} diagonal element of $\mathbf{D}$ and $\langle \mathbf{U}_j, \mathbf{Y} \rangle$ is the inner product between $\mathbf{U}_j$ and $\mathbf{Y}$ and $\mathbf{U}_j$ ($\mathbf{V}_j$ are respectively the j^{th} column of $\mathbf{U}$ and $\mathbf{V}$). In particular, when $\lambda = 0$, ridge regression reduces to OLS and we have

$$\hat{\boldsymbol{\beta}}^{\text{OLS}} = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}^T\mathbf{Y} = \sum_{j=1}^p \frac{1}{d_j} \langle \mathbf{U}_j, \mathbf{Y} \rangle \mathbf{V}_j. \tag{2.41}$$

Based on (2.40) we suggest the following procedure to compute ridge regression at a fine grid $\lambda_1, \dots, \lambda_M$:

1. Compute the SVD of $\mathbf{X}$ and save $\mathbf{U}, \mathbf{D}, \mathbf{V}$.
2. Compute $\mathbf{w}_j = \frac{1}{d_j} \langle \mathbf{U}_j, \mathbf{Y} \rangle \mathbf{V}_j$ for $j = 1, \dots, p$ and save $\mathbf{w}_j$ s.
3. For $m = 1, 2, \dots, M$,

- (i). compute $\gamma_j = \frac{d_j^2}{d_j^2 + \lambda_m}$

- (ii). compute $\hat{\boldsymbol{\beta}}_{\lambda_m} = \sum_{j=1}^p \gamma_j \mathbf{w}_j$.

The essence of the above algorithm is to compute the common vectors $\{\mathbf{w}_j\}_{j=1}^p$ first and then utilize (2.40).

2.6.5 Kernel ridge regression

In this section we introduce a nonparametric generalization of ridge regression. Our discussion begins with the following theorem.

Theorem 2.4 *Ridge regression estimator is equal to*

$$\hat{\beta}_\lambda = \mathbf{X}^T (\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I})^{-1} \mathbf{Y} \quad (2.42)$$

and the fitted value of Y at $\mathbf{x}$ is

$$\hat{y} = \mathbf{x}^T \hat{\beta}_\lambda = \mathbf{x}^T \mathbf{X}^T (\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I})^{-1} \mathbf{Y} \quad (2.43)$$

Proof. Observe the following identity

$$(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}) \mathbf{X}^T = \mathbf{X}^T \mathbf{X} \mathbf{X}^T + \lambda \mathbf{X}^T = \mathbf{X}^T (\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I}).$$

Thus, we have

$$\mathbf{X}^T = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T (\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I})$$

and

$$\mathbf{X}^T (\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I})^{-1} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T.$$

Then by using (2.27) we obtain (2.42) and hence (2.43). ■

It is important to see that $\mathbf{X}\mathbf{X}^T$ and not $\mathbf{X}^T \mathbf{X}$ appears in the expression for $\hat{\beta}_\lambda$. Note that $\mathbf{X}\mathbf{X}^T$ is a $n \times n$ matrix and its ij elements are $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$. Similarly, $\mathbf{x}^T \mathbf{X}^T$ is an n -dimensional vector with the i th element being $\langle \mathbf{x}, \mathbf{x}_i \rangle$ $i = 1, \dots, n$. Therefore, the prediction by ridge regression boils down to computing the inner product between p -dimensional covariate vectors. This is the foundation of the so-called “*kernel trick*”.

Suppose that we use another “inner product” to replace the usual inner product in Theorem 2.4; then we may end up with a new ridge regression estimator. To be more specific, let us replace $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ with $K(\mathbf{x}_i, \mathbf{x}_j)$ where $K(\cdot, \cdot)$ is a known function:

$$\mathbf{x}^T \mathbf{X}^T \rightarrow (K(\mathbf{x}, \mathbf{X}_1), \dots, K(\mathbf{x}, \mathbf{X}_n)),$$

$$\mathbf{X}\mathbf{X}^T \rightarrow \mathbf{K} = (K(\mathbf{X}_i, \mathbf{X}_j))_{1 \leq i, j \leq n}.$$

By doing so, we turn (2.43) into

$$\hat{y} = (K(\mathbf{x}, \mathbf{X}_1), \dots, K(\mathbf{x}, \mathbf{X}_n)) (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{Y} = \sum_{i=1}^n \hat{\alpha}_i K(\mathbf{x}, \mathbf{X}_i), \quad (2.44)$$

where $\hat{\alpha} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{Y}$. In particular, the fitted $\mathbf{Y}$ vector is

$$\hat{\mathbf{Y}} = \mathbf{K}(\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{Y}. \quad (2.45)$$

The above formula gives the so-called kernel ridge regression. Because $\mathbf{X}\mathbf{X}^T$ is at least positive semi-definite, it is required that $\mathbf{K}$ is also positive semi-definite. Some widely used kernel functions (Hastie, Tibshirani and Friedman, 2009) include

- *linear kernel*: $K(\mathbf{x}_i, \mathbf{x}_j) = \langle \mathbf{x}_i, \mathbf{x}_j \rangle$,
- *polynomial kernel*: $K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \langle \mathbf{x}_i, \mathbf{x}_j \rangle)^d$, $d = 2, 3, \dots$,
- *radial basis kernel*: $K(\mathbf{x}_i, \mathbf{x}_j) = e^{-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2}$, $\gamma > 0$, which is the *Gaussian kernel*, and $K(\mathbf{x}_i, \mathbf{x}_j) = e^{-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|}$, $\gamma > 0$, which is the *Laplacian kernel*.

To show how we get (2.45) more formally, let us consider to approximate the multivariate regression by using the kernel basis functions $\{K(\cdot, \mathbf{x}_j)\}_{j=1}^n$ so that our observed data are now modeled as

$$Y_i = \sum_{j=1}^n \alpha_j K(\mathbf{X}_i, \mathbf{X}_j) + \varepsilon_i$$

or in matrix form $\mathbf{Y} = \mathbf{K}\boldsymbol{\alpha} + \boldsymbol{\varepsilon}$. If we apply the ridge regression

$$\frac{1}{2} \|\mathbf{Y} - \mathbf{K}\boldsymbol{\alpha}\|^2 + \lambda \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha},$$

the minimizer of the above problem is

$$\hat{\boldsymbol{\alpha}} = (\mathbf{K}^T \mathbf{K} + \lambda \mathbf{K})^{-1} \mathbf{K}^T \mathbf{Y} = \{\mathbf{K}(\mathbf{K} + \lambda \mathbf{I})\}^{-1} \mathbf{K} \mathbf{Y},$$

where we use the fact that $\mathbf{K}$ is symmetric. Assuming $\mathbf{K}$ is invertible, we easily get (2.45).

So far we have only derived the kernel ridge regression based on heuristics and the kernel trick. In Sec. 2.7 we show that the kernel ridge regression can be formally derived based on the theory of function estimation in a reproducing kernel Hilbert space.

2.7 Regression in Reproducing Kernel Hilbert Space

A *Hilbert space* is an abstract vector space endowed by the structure of an inner product. Let $\mathcal{X}$ be an arbitrary set and $\mathcal{H}$ be a Hilbert space of real-valued functions on $\mathcal{X}$, endowed by the inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$. The evaluation functional over the Hilbert space of functions $\mathcal{H}$ is a linear functional that evaluates each function at a point x :

$$L_x : f \rightarrow f(x), \forall f \in \mathcal{H}.$$

A Hilbert space $\mathcal{H}$ is called a *reproducing kernel Hilbert space* (RKHS) if, for all $x \in \mathcal{X}$, the map L_x is continuous at any $f \in \mathcal{H}$, namely, there exists some $C > 0$ such that

$$|L_x(f)| = |f(x)| \leq C \|f\|_{\mathcal{H}}, \quad \forall f \in \mathcal{H}.$$

By the Riesz representation theorem, for all $x \in \mathcal{X}$, there exists a unique element $K_x \in \mathcal{H}$ with the reproducing property

$$f(x) = L_x(f) = \langle f, K_x \rangle_{\mathcal{H}}, \quad \forall f \in \mathcal{H}.$$

Since K_x is itself a function in $\mathcal{H}$, it holds that for every $x' \in \mathcal{X}$, there exists a $K_{x'} \in \mathcal{H}$ such that

$$K_x(x') = \langle K_x, K_{x'} \rangle_{\mathcal{H}}.$$

This allows us to define the *reproducing kernel* $K(x, x') = \langle K_x, K_{x'} \rangle_{\mathcal{H}}$. From the definition, it is easy to see that the reproducing kernel K is a symmetric and semi-positive function:

$$\sum_{i,j=1}^n c_i c_j K(x_i, x_j) = \sum_{i,j=1}^n c_i c_j \langle K_{x_i}, K_{x_j} \rangle_{\mathcal{H}} = \left\| \sum_{i=1}^n c_i K_{x_i} \right\|_{\mathcal{H}}^2 \geq 0,$$

for all c 's and x 's. The reproducing Hilbert space is a class of *nonparametric functions*, satisfying the above properties.

Let $\mathcal{H}_K$ denote the reproducing kernel Hilbert space (RKHS) with kernel $K(\mathbf{x}, \mathbf{x}')$ (Wahba, 1990; Halmos, 2017). Then, the kernel $K(\mathbf{x}, \mathbf{x}')$ admits the eigen-decomposition

$$K(\mathbf{x}, \mathbf{x}') = \sum_{j=1}^{\infty} \gamma_j \psi_j(\mathbf{x}) \psi_j(\mathbf{x}'). \quad (2.46)$$

where $\gamma_j \geq 0$ are eigen-values and $\sum_{j=1}^{\infty} \gamma_j^2 < \infty$. Let g and g' be any two functions in $\mathcal{H}_K$ with expansions in terms of these eigen-functions

$$g(\mathbf{x}) = \sum_{j=1}^{\infty} \beta_j \psi_j(\mathbf{x}), \quad g'(\mathbf{x}) = \sum_{j=1}^{\infty} \beta'_j \psi_j(\mathbf{x})$$

and their inner product is defined as

$$\langle g, g' \rangle_{\mathcal{H}_K} = \sum_{j=1}^{\infty} \frac{\beta_j \beta'_j}{\gamma_j}. \quad (2.47)$$

The functional ℓ_2 norm of $g(\mathbf{x})$ is equal to

$$\|g\|_{\mathcal{H}_K}^2 = \langle g, g \rangle_{\mathcal{H}_K} = \sum_{j=1}^{\infty} \frac{\beta_j^2}{\gamma_j}. \quad (2.48)$$

The first property shows the *reproducibility* of the kernel K .

Theorem 2.5 *Let g be a function in $\mathcal{H}_K$. The following identities hold:*

- (i). $\langle K(\cdot, \mathbf{x}'), g \rangle_{\mathcal{H}_K} = g(\mathbf{x}')$,
- (ii). $\langle K(\cdot, \mathbf{x}_1), K(\cdot, \mathbf{x}_2) \rangle_{\mathcal{H}_K} = K(\mathbf{x}_1, \mathbf{x}_2)$.
- (iii). If $g(\mathbf{x}) = \sum_{i=1}^n \alpha_i K(\mathbf{x}, \mathbf{x}_i)$, then $\|g\|_{\mathcal{H}_K}^2 = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j)$.

Proof. Write $g(\mathbf{x}) = \sum_{j=1}^{\infty} \beta_j \psi_j(\mathbf{x})$, by (2.46) we have $K(\mathbf{x}, \mathbf{x}') = \sum_{j=1}^{\infty} (\gamma_j \psi_j(\mathbf{x}')) \psi_j(\mathbf{x})$. Thus

$$\langle K(\cdot, \mathbf{x}'), g \rangle_{\mathcal{H}_K} = \sum_{j=1}^{\infty} \frac{\beta_j \gamma_j \psi_j(\mathbf{x}')}{\gamma_j} = \sum_{j=1}^{\infty} \beta_j \psi_j(\mathbf{x}') = g(\mathbf{x}').$$

This proves part (i). Now we apply part (i) to get part (ii) by letting $g(\mathbf{x}) = K(\mathbf{x}, \mathbf{x}_2)$.

For part (iii) we observe that

$$\begin{aligned} \|g\|_{\mathcal{H}_K}^2 &= \left\langle \sum_{i=1}^n \alpha_i K(\mathbf{x}, \mathbf{x}_i), \sum_{j=1}^n \alpha_j K(\mathbf{x}, \mathbf{x}_j) \right\rangle_{\mathcal{H}_K} \\ &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \langle K(\mathbf{x}, \mathbf{x}_i), K(\mathbf{x}, \mathbf{x}_j) \rangle_{\mathcal{H}_K} \\ &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j), \end{aligned}$$

where we have used part (ii) in the final step. ■

Consider a general regression model

$$Y = f(\mathbf{X}) + \varepsilon \quad (2.49)$$

where ε is independent of $\mathbf{X}$ and has zero mean and variance σ^2 . Given a realization $\{(\mathbf{X}_i, Y_i)\}_{i=1}^n$ from the above model, we wish to fit the regression function in $\mathcal{H}_K$ via the following penalized least-squares:

$$\hat{f} = \operatorname{argmin}_{f \in \mathcal{H}_K} \sum_{i=1}^n [Y_i - f(\mathbf{X}_i)]^2 + \lambda \|f\|_{\mathcal{H}_K}^2, \quad \lambda > 0. \quad (2.50)$$

Note that without the $\|f\|_{\mathcal{H}_K}^2$ term there are infinitely many functions in $\mathcal{H}_K$ that can fit the observations perfectly, i.e., $Y_i = f(\mathbf{X}_i)$ for $i = 1, \dots, n$. By using the eigen-function expansion of f

$$f(\mathbf{x}) = \sum_{j=1}^{\infty} \beta_j \psi_j(\mathbf{x}), \quad (2.51)$$

an equivalent formulation of (2.50) is

$$\min_{\{\beta_j\}_{j=1}^{\infty}} \sum_{i=1}^n [Y_i - \sum_{j=1}^{\infty} \beta_j \psi_j(\mathbf{X}_i)]^2 + \lambda \sum_{j=1}^{\infty} \frac{1}{\gamma_j} \beta_j^2. \quad (2.52)$$

Define $\beta_j^* = \frac{\beta_j}{\sqrt{\gamma_j}}$ and $\psi_j^* = \sqrt{\gamma_j} \psi_j$ for $j = 1, 2, \dots$. Then (2.52) can be rewritten as

$$\min_{\{\beta_j^*\}_{j=1}^{\infty}} \sum_{i=1}^n [Y_i - \sum_{j=1}^{\infty} \beta_j^* \psi_j^*(\mathbf{X}_i)]^2 + \lambda \sum_{j=1}^{\infty} (\beta_j^*)^2. \quad (2.53)$$

The above can be seen as a ridge regression estimate in an infinite dimensional

space. Symbolically, our covariate vector is now $(\psi_1^*(\mathbf{x}), \psi_2^*(\mathbf{x}), \dots)$ and the enlarged design matrix is

$$\mathbf{\Psi} = \begin{pmatrix} \psi_1^*(\mathbf{X}_1) & \cdots & \psi_j^*(\mathbf{X}_1) & \cdots \\ \vdots & \cdots & \vdots & \cdots \\ \psi_1^*(\mathbf{X}_n) & \cdots & \psi_j^*(\mathbf{X}_n) & \cdots \end{pmatrix}.$$

Because Theorem 2.4 is valid for any finite dimensional covariate space, it is not unreasonable to extrapolate it to the above infinite dimensional setting. The key assumption is that we can compute the inner product in the enlarged space. This is indeed true because

$$\text{inner product} = \sum_{j=1}^{\infty} \psi_j^*(\mathbf{x}_i) \psi_j^*(\mathbf{x}_{i'}) = \sum_{j=1}^{\infty} \gamma_j \psi_j(\mathbf{x}_i) \psi_j(\mathbf{x}_{i'}) = K(\mathbf{x}_i, \mathbf{x}_{i'}).$$

Now we can directly apply the kernel ridge regression formula from Section 2.6.5 to get

$$\hat{f}(\mathbf{x}) = \sum_{i=1}^n \hat{\alpha}_i K(\mathbf{x}, \mathbf{X}_i), \quad (2.54)$$

where $\mathbf{K} = (K(\mathbf{X}_i, \mathbf{X}_j))_{1 \leq i, j \leq n}$ and

$$\hat{\boldsymbol{\alpha}} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{Y}. \quad (2.55)$$

We have derived (2.54) by extrapolating Theorem 2.4 to an infinite dimensional space. Although the idea seems correct, we still need a rigorous proof. Moreover, Theorem 2.4 only concerns ridge regression, but it turns out that (2.54) can be made much more general.

Theorem 2.6 *Consider a general loss function $L(y, f(\mathbf{x}))$ and let*

$$\hat{f} = \operatorname{argmin}_{f \in \mathcal{H}_K} \sum_{i=1}^n L(y_i, f(\mathbf{x}_i)) + P_\lambda(\|f\|_{\mathcal{H}_K}), \quad \lambda > 0.$$

where $P_\lambda(t)$ is a strictly increasing function on $[0, \infty)$. Then we must have

$$\hat{f}(\mathbf{x}) = \sum_{i=1}^n \hat{\alpha}_i K(\mathbf{x}, \mathbf{X}_i) \quad (2.56)$$

where $\hat{\boldsymbol{\alpha}} = (\hat{\alpha}_1, \dots, \hat{\alpha}_n)$ is the solution to the following problem

$$\min_{\boldsymbol{\alpha}} \sum_{i=1}^n L\left(y_i, \sum_{j=1}^n \alpha_j K(\mathbf{x}, \mathbf{x}_j)\right) + P_\lambda(\sqrt{\boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha}}). \quad (2.57)$$

Proof. Any function f in $\mathcal{H}_K$ can be decomposed as the sum of two functions: one is in the span $\{K(\cdot, \mathbf{X}_1), \dots, K(\cdot, \mathbf{X}_n)\}$ and the other is in the orthogonal complement. In other words, we write

$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i K(\mathbf{x}, \mathbf{X}_i) + r(\mathbf{x})$$

where $\langle r(\mathbf{x}), K(\mathbf{x}, \mathbf{X}_i) \rangle_{\mathcal{H}_K} = 0$ for all $i = 1, 2, \dots, n$. By part (i) of Theorem 2.5 we have

$$r(\mathbf{x}_i) = \langle r, K(\cdot, \mathbf{X}_i) \rangle_{\mathcal{H}_K} = 0, \quad 1 \leq i \leq n.$$

Denote by $g(\mathbf{x}) = \sum_{i=1}^n \alpha_i K(\mathbf{x}, \mathbf{X}_i)$. Then we have $g(\mathbf{X}_i) = f(\mathbf{X}_i)$ for all i , which implies

$$\sum_{i=1}^n L(Y_i, f(\mathbf{X}_i)) = \sum_{i=1}^n L(Y_i, g(\mathbf{X}_i)). \quad (2.58)$$

Moreover, we notice

$$\begin{aligned} \|f\|_{\mathcal{H}_K}^2 &= \langle g + r, g + r \rangle_{\mathcal{H}_K} \\ &= \langle g, g \rangle_{\mathcal{H}_K} + \langle r, r \rangle_{\mathcal{H}_K} + 2\langle g, r \rangle_{\mathcal{H}_K} \end{aligned}$$

and

$$\langle g, r \rangle_{\mathcal{H}_K} = \sum_{i=1}^n \alpha_i \langle K(\cdot, \mathbf{X}_i), r \rangle_{\mathcal{H}_K} = 0.$$

Thus $\|f\|_{\mathcal{H}_K}^2 = \|g\|_{\mathcal{H}_K}^2 + \|r\|_{\mathcal{H}_K}^2$. Because $P_\lambda(\cdot)$ is a strictly increasing function, we then have

$$P_\lambda(\|f\|_{\mathcal{H}_K}) \geq P_\lambda(\|g\|_{\mathcal{H}_K}) \quad (2.59)$$

and the equality holds if and only if $f = g$. Combining (2.58) and (2.59) we prove (2.56).

To prove (2.57), we use (2.56) and part (iii) of Theorem 2.5 to write

$$\|f\|_{\mathcal{H}_K}^2 = \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha}. \quad (2.60)$$

Hence $P_\lambda(\|f\|_{\mathcal{H}_K}) = P_\lambda(\sqrt{\boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha}})$ under (2.56). ■

Theorem 2.6 is known as the *representer theorem* (Wahba, 1990). It shows that for a wide class of statistical estimation problems in a RKHS, although the criterion is defined in an infinite dimensional space, the solution always has a finite dimensional representation based on the kernel functions. This provides a solid mathematical foundation for the kernel trick without resorting to any optimization/computational arguments.

Let the loss function in Theorem 2.6 be the squared error loss and $P_\lambda(t) = \lambda t^2$. Then Theorem 2.6 handles the problem defined in (2.50) and (2.57) reduces to

$$\min_{\boldsymbol{\alpha}} \|\mathbf{Y} - \mathbf{K} \boldsymbol{\alpha}\|^2 + \lambda \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha}. \quad (2.61)$$

Table 2.1: A list of commonly used kernels.

Linear kernel	$K(\mathbf{x}_i, \mathbf{x}_j) = \langle \mathbf{x}_i, \mathbf{x}_j \rangle$
Polynomial kernel	$K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \langle \mathbf{x}_i, \mathbf{x}_j \rangle)^d$
Gaussian kernel	$K(\mathbf{x}_i, \mathbf{x}_j) = e^{-\gamma \ \mathbf{x}_i - \mathbf{x}_j\ ^2}$
Laplacian kernel	$K(\mathbf{x}_i, \mathbf{x}_j) = e^{-\gamma \ \mathbf{x}_i - \mathbf{x}_j\ }$

It is easy to see the solution is

$$\hat{\boldsymbol{\alpha}} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{Y}.$$

which is identical to (2.55). The fitted multivariate nonparametric regression function is given by (2.56). In practice, one takes a kernel function from the list of linear, polynomial, Gaussian or Laplacian kernels given in Table 2.1. It remains to show how to choose the regularization parameter λ (and γ for Gaussian and Laplacian kernels) to optimize the prediction performance. This can be done by cross-validation methods outlined in the next section.

2.8 Leave-one-out and Generalized Cross-validation

We have seen that both ridge regression and the kernel ridge regression use a *tuning parameter* λ . In practice, we would like to use the data to pick a data-driven λ in order to achieve the “best” estimation/prediction performance. This problem is often called tuning parameter selection and is ubiquitous in modern statistics and machine learning. A general solution is *k-fold cross-validation* (CV), such as 10-fold or 5-fold CV. *k-fold CV* estimates prediction errors as follows.

- Divide data randomly and evenly into k subsets.
- Use one of the subsets as the *testing set* and the remaining $k - 1$ subsets of data as a *training set* to compute testing errors.
- Compute testing errors for each of k subsets of data and average these testing errors.

An interesting special case is the *n-fold CV*, which is also known as the *leave-one-out CV*.

In this section we focus on regression problems under the squared error loss. Following the above scheme, the leave-one-out CV error, using the quadratic loss, is defined as

$$\text{CV} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{f}^{(-i)}(\mathbf{X}_i))^2, \quad (2.62)$$

where $\hat{f}^{(-i)}(\mathbf{X}_i)$ is the predicted value at $\mathbf{x}_i$ computed by using all the data except the i th observation. So in principle we need to repeat the same data

Table 2.2: A list of commonly used regression methods and their $\mathbf{S}$ matrices. d_j s are the singular values of $\mathbf{X}$ and γ_i s are the eigenvalues of $\mathbf{K}$.

Method	$\mathbf{S}$	$\text{tr } \mathbf{S}$
Multiple Linear Regression	$\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$	p
Ridge Regression	$\mathbf{X}(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T$	$\sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda}$
Kernel Regression in RKHS	$\mathbf{K}(\mathbf{K} + \lambda \mathbf{I})^{-1}$	$\sum_{i=1}^n \frac{\gamma_i}{\gamma_i + \lambda}$

fitting process n times to compute the leave-one-out CV. Fortunately, we can avoid much computation for many popular regression methods.

A fitting method is called a *linear smoother* if we can write

$$\hat{\mathbf{Y}} = \mathbf{S}\mathbf{Y} \quad (2.63)$$

for any dataset $\{(\mathbf{X}_i, Y_i)\}_{i=1}^n$ where $\mathbf{S}$ is a $n \times n$ matrix that only depends on $\mathbf{X}$. Many regression methods are linear smoothers with different $\mathbf{S}$ matrices. See Table 2.2.

Assume that a linear smoother is fitted on $\{\mathbf{X}_i, Y_i\}_{i=1}^n$. Let $\mathbf{x}$ be a new co-variate vector and $\hat{f}(\mathbf{x})$ be its the predicted value by using the linear smoother. We then augment the dataset by including $(\mathbf{x}, \hat{f}(\mathbf{x}))$ and refit the linear smoother on this augmented dataset. The linear smoother is said to be *self-stable* if the fit based on the augmented dataset is identical to the fit based on the original data regardless of $\mathbf{x}$.

It is easy to check that the three linear smoothers in Table 2.2 all have the self-stable property.

Theorem 2.7 *For a linear smoother $\hat{\mathbf{Y}} = \mathbf{S}\mathbf{Y}$ with the self-stable property, we have*

$$Y_i - \hat{f}^{(-i)}(\mathbf{X}_i) = \frac{Y_i - \hat{Y}_i}{1 - S_{ii}}, \quad (2.64)$$

and its leave-one-out CV error is equal to $\frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i - \hat{Y}_i}{1 - S_{ii}} \right)^2$.

Proof. We first apply the linear smoother to all the data except the i th to compute $\hat{f}^{(-i)}(\mathbf{X}_i)$. Write $\tilde{y}_j = y_j$ for $j \neq i$ and $\tilde{y}_i = \hat{f}^{(-i)}(\mathbf{X}_i)$. Then we apply the linear smoother to the following working dataset:

$$\{(\mathbf{X}_j, Y_j), j \neq i, (\mathbf{X}_i, \tilde{Y}_i)\}$$

The self-stable property implies that the fit stays the same. In particular,

$$\tilde{Y}_i = \hat{f}^{(-i)}(\mathbf{X}_i) = (\mathbf{S}\tilde{\mathbf{Y}})_i = S_{ii}\tilde{Y}_i + \sum_{j \neq i} S_{ij}Y_j \quad (2.65)$$

and

$$\hat{Y}_i = (\mathbf{S}\mathbf{Y})_i = S_{ii}Y_i + \sum_{j \neq i} S_{ij}Y_j. \quad (2.66)$$

Combining (2.65) and (2.66) yields

$$\tilde{Y}_i = \frac{\hat{Y}_i - S_{ii}Y_i}{1 - S_{ii}}.$$

Thus,

$$Y_i - \tilde{Y}_i = Y_i - \frac{\hat{Y}_i - S_{ii}Y_i}{1 - S_{ii}} = \frac{Y_i - \hat{Y}_i}{1 - S_{ii}}.$$

The proof is now complete. ■

Theorem 2.7 shows a nice shortcut for computing the leave-one-out CV error of a self-stable linear smoother. For some smoothers $\text{tr } \mathbf{S}$ can be computed more easily than its diagonal elements. To take advantage of this, *generalized cross-validation* (GCV) (Golub, Heath and Wahba, 1979) is a convenient computational approximation of the leave-one-out CV error. Suppose that we approximate each diagonal elements of $\mathbf{S}$ by their average which equals $\frac{\text{tr } \mathbf{S}}{n}$, then we have

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i - \hat{Y}_i}{1 - S_{ii}} \right)^2 \approx \frac{1}{n} \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{(1 - \frac{\text{tr } \mathbf{S}}{n})^2} := \text{GCV}.$$

In the literature $\text{tr } \mathbf{S}$ is called the *effective degrees of freedom* of the linear smoother. Its rigorous justification is based on Stein's unbiased risk estimation theory (Stein, 1981; Efron, 1986). In Table 2.2 we list the degrees of freedom of three popular linear smoothers.

Now we are ready to handle the tuning parameter selection issue in the linear smoother. We write $\mathbf{S} = \mathbf{S}_\lambda$ and

$$\text{GCV}(\lambda) = \frac{1}{n} \frac{\mathbf{Y}^T (\mathbf{I} - \mathbf{S}_\lambda)^2 \mathbf{Y}}{(1 - \frac{\text{tr } \mathbf{S}_\lambda}{n})^2}.$$

According to GCV, the best λ is given by

$$\lambda^{\text{GCV}} = \underset{\lambda}{\text{argmin}} \frac{1}{n} \frac{\mathbf{Y}^T (\mathbf{I} - \mathbf{S}_\lambda)^2 \mathbf{Y}}{(1 - \frac{\text{tr } \mathbf{S}_\lambda}{n})^2}.$$

2.9 Exercises

2.1 Suppose that $\{\mathbf{X}_i, Y_i\}$, $i = 1, \dots, n$ is a random sample from linear regression model (2.1). Assume that the random error $\varepsilon \sim N(0, \sigma^2)$ and is independent of $\mathbf{X} = (X_1, \dots, X_p)^T$.

- (a) Show that the maximum likelihood estimate of β is the same as its least squares estimator, while the maximum likelihood estimate of σ^2 is RSS/n , where $\text{RSS} = \sum_{i=1}^n (Y_i - \mathbf{X}_i^T \hat{\beta})^2$ is the residual sum-of-squares and $\hat{\beta}$ is the least squares estimator.
- (b) Assume that $\mathbf{X}$ is of full rank. Show that $\text{RSS} \sim \sigma^2 \chi_{n-p}^2$.
- (c) Prove that $1 - \alpha$ CI for β_j is $\hat{\beta}_j \pm t_{n-p}(1 - \alpha/2) \sqrt{v_j \text{RSS}/(n-p)}$, where v_j is the j^{th} diagonal element of $(\mathbf{X}^T \mathbf{X})^{-1}$.
- (d) Dropping the normality assumption, if $\{\mathbf{X}_i\}$ are independent and identically distributed from a population with $E \mathbf{X}_1 \mathbf{X}_1^T = \Sigma$ and independent of $\{\varepsilon_i\}_{i=1}^n$, which are independent and identically distributed from a population with $E \varepsilon = 0$ and $\text{Var}(\varepsilon) = \sigma^2$, show that

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 \Sigma^{-1}).$$

2.2 Suppose that a random sample of size n from linear regression model (2.1), where the random error $\varepsilon \sim N(0, \sigma^2)$ and is independent of $(X_1, \dots, X_p)$. Consider a general linear hypothesis $H_0 : \mathbf{C}\beta = \mathbf{h}$ versus $H_0 : \mathbf{C}\beta \neq \mathbf{h}$, where $\mathbf{C}$ is a $q \times p$ constant matrix with rank q ($\leq p$), and $\mathbf{h}$ is a $q \times 1$ constant vector.

- (a) Derive the least squares estimator of β under H_0 , denoted by $\hat{\beta}_0$.
- (b) Define $\text{RSS}_1 = \|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2$ and $\text{RSS}_0 = \|\mathbf{Y} - \mathbf{X}\hat{\beta}_0\|^2$, the residual sum-of-squares under H_1 and H_0 . Show that $\text{RSS}_1/\sigma^2 \sim \chi_{n-p}^2$. Further, under the null hypothesis H_0 , $(\text{RSS}_0 - \text{RSS}_1)/\sigma^2 \sim \chi_q^2$ and is independent of RSS_1 .
- (c) Show that under H_0 , $F = \{(\text{RSS}_0 - \text{RSS}_1)/q\}/\{\text{RSS}_1/(n-p)\}$ follows an $F_{q, n-p}$ distribution.
- (d) Show that the F -test for H_0 is equivalent to the likelihood ratio test for H_0 .

2.3 Suppose that we have n independent data $Y_i \sim N(\mu, \sigma_i^2)$, where $\sigma_i = \sigma^2 v_i$ with known v_i . Use the weighted least-squares method to find an estimator of μ . Show that it is the best linear unbiased estimator. Compare the variance of the sample mean $\bar{y}$ with that of the weighted least-squares estimator $v_i^2 = \log(i+1)$ when $n = 20$.

2.4 Consider the linear model $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$, where $\varepsilon \sim N(0, \Sigma)$, and $\mathbf{X}$ is of full rank.

- (a) Show that the general least-squares estimator, which minimizes $(\mathbf{Y} - \mathbf{X}\beta)^T \Sigma^{-1} (\mathbf{Y} - \mathbf{X}\beta)$, is the best linear unbiased estimator. More precisely, for any vector $\mathbf{c} \neq 0$, $\mathbf{c}^T \hat{\beta}$ is the best linear estimator of β . Do we need the normality assumption?
- (b) Deduce from part (a) that the weighted least-squares estimator is the best linear unbiased estimator, when the error distribution is uncorrelated.

- (c) If Σ is the equi-correlation matrix with unknown correlation ρ , what is the solution to part (a)?

2.5 Suppose that $Y_1, \dots, Y_n$ are random variables with common mean μ and covariance matrix $\sigma^2 \mathbf{V}$, where $\mathbf{V}$ is of the form $v_{ii} = 1$ and $v_{ij} = \rho$ ($0 < \rho < 1$) for $i \neq j$.

- (a) Find the generalized least squares estimate of μ .
 (b) Show that it is the same as the ordinary least squares estimate.

2.6 Suppose that data $\{X_{i1}, \dots, X_{ip}, Y_i\}, i = 1, \dots, n$, are an independent and identically distributed sample from the model

$$Y = f(X_1\beta_1 + \dots + X_p\beta_p + \varepsilon),$$

where $\varepsilon \sim N(0, \sigma^2)$ with unknown σ^2 , and $f(\cdot)$ is a known, differentiable, strictly increasing, non-linear function.

- (a) Consider transform $Y_i^* = h(Y_i)$, where $h(\cdot)$ is a differentiable function yet to be determined. Show that $\text{Var}(Y_i^*) = \text{constant}$ for all i leads to the equation: $[h'\{f(u)\}]^2 \{f'(u)\}^2 = \text{constant}$ for all u .
 (b) Let $f(x) = x^p$ ($p > 1$). Find the corresponding $h(\cdot)$ using the equation in (a).
 (c) Let $f(x) = \exp(x)$. Find the corresponding h transform.

2.7 The data set 'hkepd.txt' consists of daily measurements of levels of air pollutants and the number of total hospital admissions for circulatory and respiratory problems from January 1, 1994 to December 31, 1995 in Hong Kong. This data set can be downloaded from this book website. Of interest is to investigate the association between the number of total hospital admissions and the levels of air pollutants.

We set the Y variable to be the number of total hospital admissions and the X variables the levels of air pollutants. Define

X_1 = the level of sulfur dioxide ($\mu g/m^3$);
 X_2 = the level of nitrogen dioxide ($\mu g/m^3$);
 X_3 = the level of dust ($\mu g/m^3$).

- (a) Fit the data to the following linear regression model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon, \quad (2.67)$$

and test whether the level of each air pollutant has significant impact on the number of total hospital admissions.

- (b) Construct residual plots and examine whether the random error approximately follows a normal distribution.
 (c) Take $Z = \log(Y)$ and fit the data to the following linear regression model

$$Z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon, \quad (2.68)$$

and test whether the level of each air pollutant has significant impact on the logarithm of the number of total hospital admissions.

- (d) Construct residual plots based on model (2.68) and compare this residual plot with the one obtained in part (b).
- (e) Since the observations in this data set were collected over time, it is of interest to explore the potential seasonal trends in the total number of hospital admissions. For simplicity, define t to be the day on which data were collected. This corresponds to the first column of the data set. Consider the time-varying effect model

$$Z = \beta_0(t) + \beta_1(t)X_1 + \beta_2(t)X_2 + \beta_3(t)X_3 + \varepsilon, \quad (2.69)$$

which allows the effects of predictors varying over time. Model (2.69) indeed is a nonparametric regression model. Fit model (2.69) to the data by using nonparametric regression techniques introduced in Section 2.5.

- (f) For model (2.69), construct an F -type test for $H_0 : \beta_3(\cdot) \equiv 0$ (i.e., the level of dust not significant) by comparing their residual sum of squares under H_0 and H_1 .

2.8 The data set ‘macroecno.txt’ consists of 129 macroeconomic time series and can be downloaded from this book website. Let the response $Y_t = \Delta \log(\text{PCE}_t)$ be the changes in personal consumption expenditure. Define covariates as follows.

$$\begin{aligned} X_{t,1} &= \text{Unrate}_{t-1}, \quad X_{t,2} = \Delta \log(\text{IndPro}_{t-1}), \quad X_{t,3} = \Delta \log(\text{M2Real}_{t-1}), \\ X_{t,4} &= \Delta \log(\text{CPI}_{t-1}), \quad X_{t,5} = \Delta \log(\text{SPY}_{t-1}), \quad X_{t,6} = \text{HouSta}_{t-1}, \\ X_{t,7} &= \text{FedFund}_{t-1} \end{aligned}$$

Set the last 10 years data as testing data and the remaining as training data. Conduct linear regression analysis and address the following questions.

- (a) What are $\hat{\sigma}^2$, adjusted R^2 and insignificant variables?
- (b) Conduct the stepwise deletion, eliminating one least significant variable at a time (by looking at the smallest $|t|$ -statistic) until all variables are statistically significant. Name this model as model $\widehat{\mathcal{M}}$. (The function `step` can do the job automatically)
- (c) Using model $\widehat{\mathcal{M}}$, what are root mean-square prediction error and mean absolute deviation prediction error for the test sample?
- (d) Compute the standardized residuals. Present the time series plot of the residuals, fitted values versus the standardized residuals, and QQ plot for the standardized residuals.

2.9 Zillow is an online real estate database company that was founded in 2006. The most important task for Zillow is to predict the house price. However, their accuracy has been criticized a lot. According to Fortune,

“Zillow has Zestimated the value of 57 percent of U.S. housing stock, but only 65 percent of that could be considered ‘accurate’ by its definition, within 10 percent of the actual selling price. And even that accuracy isn’t equally distributed”. Therefore, Zillow needs your help to build a housing pricing model to improve their accuracy. Download the data from the book website, and read the data (training data: 15129 cases, testing data: 6484 cases)

```
train.data <- read.csv('train.data.csv', header=TRUE)
test.data <- read.csv('test.data.csv', header=TRUE)
train.data$zipcode <- as.factor(train.data$zipcode)
test.data$zipcode <- as.factor(test.data$zipcode)
```

where the last two lines make sure that zip code is treated as a factor. Let $\mathcal{T}$ be a test set, define out-of-sample R^2 as of a prediction method $\{\hat{y}_i^{pred}\}$ as

$$R^2 = 1 - \frac{\sum_{i \in \mathcal{T}} (y_i - \hat{y}_i^{pred})^2}{\sum_{i \in \mathcal{T}} (y_i - \bar{y}^{pred})^2},$$

where $\bar{y}^{pred} = \text{ave}(\{y_i\}_{i \in \mathcal{T}_0})$ and $\mathcal{T}_0$ is the training set.

- (a) Calculate out-of-sample R^2 using variables “bedrooms”, “bathrooms”, “sqft_living”, and “sqft_lot”.
- (b) Calculate out-of-sample R^2 using the 4 variables above along with interaction terms.
- (c) Compare the result with the nonparametric model using the Gaussian kernel with $\gamma = 0.3^2/2$ and $\gamma = 0.1^2/2$ (standardize predictors first) and λ chosen by 5-fold CV or GCV. **Hint:** To speed up computation, please divide data randomly into 10 pieces and get 10 predicted values based on 10 fitted kernel models. Use the median of these 10 predicted values as your final prediction.
- (d) Add the factor zipcode to (b) and compute out-of-sample R^2 .
- (e) Add the following additional variables to (d): $X_{12} = I(\text{view} == 0)$, $X_{13} = L^2$, $X_{13+i} = (L - \tau_i)_+^2$, $i = 1, \dots, 9$, where τ_i is $10 * i^{th}$ percentile and L is the size of the living area (“sqft_living”). Compute out-of-sample R^2 .
- (f) Why do you see the increased out-of-sample R^2 with modeling complexity?